

This project roadmap follows an Agile methodology, utilizing **2-week Sprints**. The project is estimated to take **24 weeks (6 months)** to reach a stable MVP (Minimum Viable Product).

1. Estimated Duration

- **Total Duration:** 24 Weeks
- **Phase 1 (Discovery & Architecture):** Weeks 1–4
- **Phase 2 (Core Development):** Weeks 5–16
- **Phase 3 (AI & Advanced Features):** Weeks 17–20
- **Phase 4 (Testing, Compliance & Deployment):** Weeks 21–24

2. Milestones

- **M1:** Architecture Approval & Security Compliance Framework (Week 4)
- **M2:** MVP Core (Auth, Accounts, Transfers) (Week 12)
- **M3:** Loan Management & Employee Workflow (Week 16)
- **M4:** AI Insights & Fraud Detection Integration (Week 20)
- **M5:** Final Security Audit & Go-Live (Week 24)

3. Weekly Timeline

| Phase | Weeks | Focus Area | Key Deliverables |

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| **Discovery** | 1-4 | Requirements refinement, System Architecture, SecOps setup. | SRS sign-off, Tech Stack finalized, Security Policy. |

| **Foundation** | 5-8 | User Auth (MFA), Account creation, DB Schema. | Functional User Login, Account Dashboard. |

| **Core Dev** | 9-12 | Transaction engine, Ledger, Internal transfers. | Transaction processing within 3s SLA. |

| **Operations** | 13-16 | Loan management portal, Employee dashboards. | Loan workflow (Pending/Approve/Reject). |

| **Intelligence** | 17-20 | Fraud detection, AI spending analytics. | Fraud flag alerts, Spending insight reports. |

| **Finalization** | 21-24 | Penetration testing, Load testing, Deployment. | Compliance report, Production-ready system. |

4. Deliverables

- **Technical:** Encrypted Database, MFA Module, Transaction Ledger, Loan Processing Logic, Fraud Detection Engine.
- **Functional:** Customer Portal (Web/Mobile responsive), Employee Admin Dashboard, Audit Logging Service.
- **Documentation:** API Documentation, Security Compliance Report (PCI-DSS/GDPR), User Training Manuals.

5. Risk Assessment & Mitigation

| Risk | Impact | Mitigation Strategy |

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| **Security Breach** | High | Implement "Security by Design," conduct weekly automated scans, and perform quarterly 3rd-party pentesting. |

| **Regulatory Change** | Medium | Maintain modular architecture to allow rapid adjustments to logic (e.g., interest rates, compliance rules). |

| **False Positive Fraud** | Medium | Implement a "Review Queue" where users can dispute flagged transactions; allow AI tuning via admin console. |

| **Scalability Bottlenecks** | Low | Utilize horizontal scaling/microservices; monitor DB performance using observability tools (Prometheus/Grafana). |

| **Integration Failure** | High | Use robust API mocking during dev; maintain strong contract testing with external credit bureaus. |

6. Agile Governance (The "PM View")

- **Sprint Reviews:** Demo functional increments to stakeholders every 2 weeks.
- **Daily Standups:** Focus on blockers regarding security compliance and data integration.
- **Retrospectives:** Analyze "False Positive" rates and system latency issues to adjust the backlog for the next sprint.
- **Definition of Done (DoD):** Feature is code-reviewed, unit-tested, security-scanned, and accepted by Product Owner.

Next Steps for Implementation:

1. **Kick-off Meeting:** Align on the specific API for Credit Bureau integration.
2. **Infrastructure Provisioning:** Set up cloud environment (AWS/Azure/GCP) with appropriate data residency settings.
3. **Backlog Grooming:** Prioritize the "Transaction Processing" backlog items for Sprint 1.